



Tatiana Eyenga Ovono

Computer Scientist · Optimization, AI Security & Secure Payment Systems

tatiana.eyenga@facsciences-uy1.cm · eyengaovono@gmail.com
+237 693 544 211 · +237 671 466 734 · Yaoundé, Cameroon

PROFESSIONAL SUMMARY

Computer scientist completing a PhD in optimization and secure payment systems at the University of Yaoundé I. Doctoral work produced **CostGA**, an evolutionary-algorithm optimizer that reduces large-batch transaction settlement time by two-thirds. Combines applied research with six years of operational and teaching experience — including managing university-wide IT infrastructure — and holds **ISO 8583 Lead Auditor** and **ISO/IEC 27001 Lead Implementer** certifications. Seeking to bring rigorous, security-focused research to international financial and technology institutions.

EDUCATION

Ph.D. in Computer Science

Nov 2020 – Present

University of Yaoundé I, Cameroon

Thesis: Heuristic Modeling and Optimization of Resource Consumption in Interorganizational Business Process Execution — Applications to Payment Systems.

- Designed an optimization model for computational and network resource consumption under strict security and reliability constraints, implemented as **CostGA**, an evolutionary-algorithm optimizer in C / UNIX.
- Achieved a **two-thirds reduction** in inter-operator fund-transfer time on transaction batches exceeding one million.
- Defense expected by June 2026; pre-defense successfully completed in February 2026.

Master II in Computer Science

2017 – 2020

University of Yaoundé I, Cameroon

Graduated September 2020 · GPA 3.53 / 4.0.

PROFESSIONAL CERTIFICATIONS

ISO 8583:2023 — Lead Auditor

Financial Transaction Card-Originated Messages · Interchange Message Specifications

Certificate No. PC-LA-ABIS-2841 — the core messaging standard underpinning card payment and interchange systems.

ISO/IEC 27001:2022 — Lead Implementer

Information Security Management Systems

Certificate No. PC-LA-ABIS-2863 — design, implementation and maintenance of organizational information-security frameworks.

PROFESSIONAL EXPERIENCE

IT Manager

Jan 2023 – Jan 2026

Inter-States Congo-Cameroon University

- Defined and led the institution's information-system strategy, aligning IT with academic and operational goals.
- Managed the university data center — high availability, security and performance — and oversaw networks, servers and enterprise applications.
- Established and regularly tested data-security, backup and disaster-recovery procedures; provided technical leadership to staff and students.

Inter-States Congo-Cameroon University

- Taught undergraduate and graduate computer science across thirteen courses; supervised over fifty master's students through research projects and thesis defenses.
- Organized academic masterclasses and supported the coordination of research events, including the ICT4AD'24 conference.

Moodle Platform Administrator

2020

Inter-States Congo-Cameroon University

Administered the institutional e-learning platform: users, roles and permissions, course content, system load and platform security.

PUBLICATIONS

1. A Genetic Algorithm for Resources Consumption Optimisation within Interorganizational Business Process Execution
T. Eyenga Ovono & R. Atsa Etoundi · Int. Journal of Engineering Sciences (IJES), Vol. 11, No. 9s · 2025

2. The Genetic Algorithm for Interoperability Cost Optimization: A Case Study of the GIMAC Business Network
T. Eyenga Ovono · AJIAM, Vol. 1, No. 1 · 2025

3. Digital Transformation Model in the State Universities in Cameroon based on Actor Network Theory
T. Eyenga Ovono, G. Abessolo Alo'o & R. Atsa Etoundi · Int. Journal of Distributed and Cloud Computing, Vol. 11, No. 1 · 2023

CONFERENCE PRESENTATIONS

1. A Proposed Model for Resource Consumption Optimization within Inter-organizational Business Process Execution
CSCE'25 — World Congress in Computer Science & Applied Computing · Las Vegas, USA · July 2025

2. A Genetic Algorithm for Resource Consumption Optimization within Interorganizational Business Process Execution
ICAMS-25 — Int. Conference on Applied Mathematics & Computer Sciences · Marseille, France · May 2025

3. Genetic Algorithm for Interoperability Cost Optimization: A Case Study of Networked Enterprises
ICT4AD'25 — Int. Conference on ICT for African Development · Yaoundé, Cameroon · Nov 2024

4. Applying Actor Network Theory in Mobile Learning: Case Study of the University of Yaoundé I [Best Paper Award]
ICT4AD'19 — Int. Conference on Information and Development · 2019

TEACHING & ACADEMIC LEADERSHIP

Six years of university teaching across thirteen graduate and undergraduate courses; supervised more than fifty master's theses.

Representative courses — ITIL 4 Foundation · Web Services Orchestration · Big Data & Business Intelligence · Business Process Management · Object Modeling (UML) · Web Programming · Object-Oriented Programming · Advanced Algorithms.

- **President, Erudit Association** — promoting the dissemination of scientific research.
- Organized PhD masterclasses on research methodology and scientific writing (50+ participants, 2025; 2022).
- Organized International Girls in ICT Day at the University of Yaoundé I; led International Women's Day 2024 activities.
- Supervisor and examiner for master's thesis defenses in computer engineering; mentored student projects showcased at the MTN Startup Competition.
- Member, ASIAM-Cameroon (African Society for Industrial and Applied Mathematics).

TECHNICAL & RESEARCH SKILLS

Programming	Python, C / C++ (advanced) · UNIX / Linux environments
Research methods	Evolutionary computation & heuristic optimization · research design · scientific writing & peer-reviewed publication · grant proposals
Domains	Secure payment systems · AI for cybersecurity · adversarial machine learning · information-systems interoperability · fraud detection
Languages	English — fluent (research, teaching, conferences, professional writing) · French — native